SIMARJOT SINGH ANAND

simaranand250@gmail.com | linkedin.com/in/simarjot-singh-anand | github.com/Simar25 | simar25.github.io

CS undergraduate building end-to-end AI systems, from edge computer vision (patent filed) to LLM-powered search over company annual reports. Seeking AI/ML and software engineering internships.

EDUCATION

Vellore Institute of Technology, Chennai *2024 – 2028*
B.Tech, Computer Science & Engineering **CGPA: 8.3 / 10**
Class XII (CBSE): 93.0% · Class X (CBSE): 91.2%

TECHNICAL SKILLS

Languages	Python, C, C++, Java, SQL, JavaScript, PHP, R
AI / ML	scikit-learn, XGBoost, SHAP, SMOTE, YOLOv8, MobileNetV3, RAG, embeddings, Gemini API
Data & Backend	pandas, NumPy, DuckDB, MySQL, ChromaDB, Flask, FastAPI, Streamlit
Web & Tools	HTML, CSS, Git, GitHub, pytest, Raspberry Pi, Kaggle

PROJECTS

Ambient Vibe Intelligence System – *Patent filed* *Raspberry Pi, YOLOv8, MobileNetV3, FastAPI*

- Co-invented an edge-AI IoT device that turns a Raspberry Pi camera feed into a real-time "vibe" score for hospitality venues by fusing lighting, occupancy, scene-aesthetic and infrastructure signals.
- Built the on-device inference pipeline: YOLOv8-nano for occupancy and MobileNetV3 for scene classification.
- Served the scores through a Flask/FastAPI REST API consumed by an artist–venue matching platform that connects restaurants with local musicians.
- Filed as a formal patent disclosure through VIT Chennai; working end-to-end prototype at TRL-3.

FilingLens: AI Analyst for SEC Annual Reports – *In progress* · GitHub *Python, DuckDB, RAG, ChromaDB, Gemini*

- Building an AI analyst that answers plain-English questions about public companies by combining retrieval over 10-K report text with SQL over financial data.
- Wrote a data pipeline that loads 10-K filings, XBRL financial statements and daily stock prices from SEC EDGAR and Yahoo Finance into DuckDB, with every figure traceable to its source filing.
- Parsed 84 annual reports into 1,862 sections and 21,005 passages, embedded them with bge-base-en-v1.5 into ChromaDB, and built a RAG pipeline that answers with numbered citations; unit-tested with pytest.

Customer Churn Prediction · Live demo · GitHub *scikit-learn, XGBoost, SHAP, Streamlit*

- Built an end-to-end churn model on the IBM Telco dataset (7,043 customers, 26% churners), applying SMOTE to the training split only to handle class imbalance without data leakage.
- Compared Logistic Regression, Random Forest and XGBoost on recall and ROC-AUC, not accuracy (best AUC 0.83).
- Explained the model with SHAP (short tenure and month-to-month contracts drive churn); live on Streamlit.

Mannsay: Online Counselling Platform – *Client project* *HTML, CSS, JavaScript, PHP, MySQL*

- Designed and delivered a responsive website for a clinical psychologist, with online appointment booking, client login and a wellness self-check.
- Built the PHP/MySQL backend for user accounts, appointments and progress tracking, plus a dashboard to manage appointment status and email notifications via EmailJS.

Multimodal Biodiversity Estimation – *Research team* *Python, XGBoost, scikit-learn*

- Worked in an AI/ML research team predicting species richness across Costa Rica from multimodal data, combining satellite imagery with climate records.
- Trained and evaluated XGBoost and scikit-learn models on the combined imagery and climate features.

ACHIEVEMENTS & CERTIFICATIONS

- Co-inventor on a patent disclosure filed through VIT Chennai (Ambient Vibe Intelligence System).
- Certifications: 19 Google Cloud Skills Boost courses (GDG on Campus VIT Chennai Study Jams, 2025); Infosys Springboard Basics of Python and Programming Fundamentals using Python (Feb 2025).
- Languages: English, Hindi, Punjabi.